

FL Simulation Results

File: test_Trimmed_Mean

Federated Learning Configuration

Total Clients (N): 10
Malicious Clients (M): 3
Neural Network: SimpleNN
Training Rounds: 15
Poisoned Rounds (R): 15
Distribution Strategy: last
Local Epochs (E): 3

Data Poisoning Attack Parameters

Poisoning Operation: Label Flip
Attack Intensity: 10.0% (0.10)
Poisoned Data Percentage: 100.0% (1.00)

Attack Summary: Using label_flip with 10.0% (0.10) intensity on 100.0% (1.00) of data for 15 rounds with 3 malicious clients.

Simulation Results

Init Accuracy: 0.7928
Clean Accuracy: 0.8699
Clean + DP Protection Accuracy: 0.8576
Poisoned Accuracy: 0.4318
Poisoned + DP Protection Accuracy: 0.5725
Drop (Clean - Poisoned): 0.4381
Drop (Clean - Init): 0.0771
Drop (Clean DP - Init): 0.0648
Drop (Poisoned - Init): -0.3610
Drop (Poisoned DP - Init): -0.2203
DP Protection Method: trimmed_mean
GPU Used: GPU 2

Confusion Matrix Metrics (Weighted Avg):

Clean Precision: 0.8760
Clean Recall: 0.8699
Clean F1 Score: 0.8691

Clean DP Precision: 0.8669
Clean DP Recall: 0.8576
Clean DP F1 Score: 0.8572

Poisoned Precision: 0.5550
Poisoned Recall: 0.4318
Poisoned F1 Score: 0.4455

DP Protection Precision: 0.7312
DP Protection Recall: 0.5725
DP Protection F1 Score: 0.5820

Summary

FL Simulation Complete
Task: b739c96b-f474-4490-9c53-240ae1ef0d01
GPU: GPU 2
Init Accuracy: 0.7928
Clean Accuracy: 0.8699
Clean DP Accuracy: 0.8576
Poisoned Accuracy: 0.4318
Data Poison Protection Accuracy: 0.5725
Drop (Clean - Poisoned): 0.4381
Drop (Clean - Init): 0.0771
Drop (Clean DP - Init): 0.0648
Drop (Poisoned - Init): -0.3610
Drop (Poisoned_DP - Init): -0.2203
Data Poison Protection Method: trimmed_mean
--- Confusion Matrix Metrics (Weighted Avg) ---
Clean Precision: 0.8760
Clean Recall: 0.8699
Clean F1 Score: 0.8691
Clean DP Precision: 0.8669
Clean DP Recall: 0.8576
Clean DP F1 Score: 0.8572
Poisoned Precision: 0.5550
Poisoned Recall: 0.4318
Poisoned F1 Score: 0.4455
DP Protection Precision: 0.7312
DP Protection Recall: 0.5725
DP Protection F1 Score: 0.5820